Binomial Expansion

Questions 1–32

Question 1. Without using your calculator, find the value of

a) $\frac{8!}{6!}$
b) $\frac{50!}{47!3!}$
c) $\frac{50!}{47!4!}$
d) $\binom{7}{6}$
e) $\binom{100}{98}$
f) $\binom{100}{99}$
g) ${}^{50}C_{49}$
h) ${}^{102}C_{102}$

Question 2. Simplify

a) $\frac{n!}{(n-1)!}$
b) $\frac{n!}{2!(n-2)!}$
c) $\binom{n}{1}$
d) $\binom{n+1}{n}$
e) ${}^{n}C_{n}$
f) $\binom{2n-1}{2}$
g) $\binom{2n-4}{2n-6}$

Question 3. Show that $10\binom{n}{5} = 21\binom{n}{3}$ simplifies to $n^2 - 7n - 30 = 0$ and hence find the value of n.

Question 4. Solve the following equations:

a) $3 \times {}^nC_6 = 11 \times {}^nC_4$

b) $28 \times {}^nC_4 = 15 \times {}^nC_6$

c) $33 \times {}^nC_2 = 2 \times {}^nC_5$

Question 5. Fully expand in ascending powers of x:

a) $(1+x)^6$
b) $(2+x)^5$
c) $(x+3)^6$
d) $(x-1)^4$

Question 6. Fully expand in descending powers of x:

a) $(x-3)^5$
b) $(5-x)^6$
c) $(2x+1)^4$
d) $(3x-3)^5$
e) $(1+\frac{1}{2}x)^4$
f) $(2-\frac{x}{4})^5$
g) $(2x-\frac{1}{x})^4$

Question 7. In ascending order, fully expand and simplify, $-5(3-2x)^4$.

Question 8. In ascending order, fully expand and simplify, $(1+2x)(2-x)^4$.

Question 9. In ascending order, fully expand and simplify, leaving your answers in exact form:

a) $(1+\sqrt{3})^4$
b) $(\sqrt{2}+\sqrt{5})^4$
c) $(\sqrt{7}-\sqrt{2})^5$

Question 10. Find the x^2 coefficient in the expansion of:

a) $(x+5)^4$
b) $(2x-1)^5$
c) $(8+x)^5$
d) $(1-3x)^6$
e) $(1+2x)^{78}$

Question 11. Find the x^2 coefficient in the expansion of:

a) $(x - \frac{1}{x})^4$

b) $(3x - \frac{2}{x})^6$

c) $(x - \frac{1}{x})^{18}$

Question 12. By using the definition of $\binom{n}{r}$, show that $\binom{n}{5} = \binom{n-1}{4} + \binom{n-1}{5}$.

Question 13. By using the definition of $\binom{n}{r}$, show that $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$.

Question 14. Explain why in the expansion of $(1-3x)^9$, the terms with odd powers of x have negative coefficients.

Question 15. Find the value of n in the expansion of $(1+x)^n$ where $n \geq 5$, given that the coefficient of x^2 and the coefficient of x^5 is the same.

Question 16. Find the value of n in the expansion of $(1+x)^n$ where $n \geq 4$, given that the coefficient of x^4 is 13 times the coefficient of x^2.

Question 17. Find the value of n in the expansion of $(1 + 2x)^n$ where $n \geq 2$, given that the coefficient of x^2 is 10 times the coefficient of x.

Question 18. Find the value of n in the expansion of $(1 + \frac{x}{2})^n$ where $n \geq 5$, given that the coefficient of x^4 is 2 times the coefficient of x^5.

Question 19. Find the x^2 coefficient in the expansion of:

a) $x^2(x + \frac{1}{x})^{12}$

b) $(1 - x)(1 + 4x)^{10}$

c) $(1 - 2x)(3 - x)^9$

Question 20.

a) Fully expand $(3 + x)^5$

b) Hence simplify $(3 - \sqrt{2})^5$ giving your answer in the form $a + b\sqrt{2}$ where a and b are constants.

Question 21.

a) Fully expand $(2+5x)^4$

b) State the full expansion of $(2-5x)^4$

c) Hence fully simplify $(2+\sqrt{75})^4+(2-\sqrt{75})^4$

Question 22.

a) Find the first 4 terms in ascending order of $(1+ax)^8$ where a is a positive constant.

b) Given that the coefficient is x^3 is 3 times the coefficient of x^2, find the value of a

Question 23. Find the possible values of a in the expansion of $(a+2x)^6$ given the x^2 term has coefficient 4860.

Question 24. Find the values of a and n such that $(1+ax)^n = 1+40x+700x^2+\cdots$ where $a, n \in \mathbb{Z}$.

Question 25. Find the value of a given in the expansion of $(2x - \frac{1}{ax})^6$ the coefficient of x^4 is -12.

Question 26.

a) In ascending order, find the expansion of $(2 - 7x)^5$

b) Given that in the expansion of $(1 + ax)(2 - 7x)^5$ the x term has coefficient of -464, find the value of a

Question 27.

a) Find the first 3 terms in ascending order of $(1-2x)^9$

b) Hence find the value of 0.98^9 giving your answer to 3 decimal places.

Question 28. Explain how you can use the expansion of $(3+\frac{2x}{5})^5$ to find the value of 2.96^5.

Question 29.

a) Find the first 3 terms in ascending order of the expansions of $(1 - \frac{x}{2})^5$ and $(2 - \frac{3x}{4})^6$

b) Hence, estimate $0.995^5 + 1.9925^6$ giving your answer to 3 decimal places.

Question 30. Without fully expanding $(x^2 + \frac{1}{x^4})^{12}$ explain why all the indices of the powers of x are multiples of 6.

Question 31.

a) Fully expand $(1+3x)^4$

b) Hence state the full expansion of $(1-3x)^4$

c) State the minimum value of $f(x) = (1+3x)^4 + (1-3x)^4$

Question 32.

a) Find the first 3 terms in ascending order in the expansion of $(5-2x)^6$

b) Hence determine the coefficient of the quadratic term in $(5-2x-6x^2)^6$

c) Use your answer to part a) to find the coefficient of the x^2 term of $(5+3x-2x^2)^6$